Water Air Soil Pollut (2014) 225:1868
DOI 10.1007/s11270-014-1868-2

Long-term Trends in Cladoceran Assemblages Related to Acidification and Subsequent Liming of Middle Lake (Sudbury, Canada)

Andrew L. Labaj · Adam Jeziorski · Joshua Kurek · John P. Smol

Received: 7 October 2013 / Accepted: 9 January 2014 / Published online: 6 February 2014

Abstract In the mid-20th century, similar to many lakes in the vicinity of Sudbury, Canada, Middle Lake was severely acidified due to nearby smelting operations. However, this lake is of particular interest because it was limed in 1973, and later fertilized as part of a restoration effort. Here, we use paleolimnological methods to track cladoceran assemblage responses to acidification, liming, and subsequent recovery in a ~250-year lake sediment record. Cladoceran assemblage changes, notably increases in *Chydorus brevilabris*, coincided with the late 1800s establishment of open-pit ore roasting in the region. As acidification progressed, the *Daphnia pulex* complex was replaced by the *Daphnia longispina* complex. At the height of acidification, and with similar timing to the liming, *C. brevilabris* increased abruptly in relative abundance in the sediment record, followed by a rapid decline. Invertebrate predation was investigated using *Bosmina* mucro length; however, no significant trends were evident. Our results suggest that complete biological recovery has not occurred. Specifically, species richness (rarefied) is ~64 % lower after the onset of acidification, and many rare species present prior to the onset of acidification have not yet returned to pre-impact levels despite dispersal events of these rare taxa being observed during contemporary zooplankton monitoring. Factors impeding the complete biological recovery of the cladocerans in Middle Lake may include biotic resistance, ongoing metal contamination, and a warming climate.

Keywords Acidification · Liming · Recovery · Zooplankton · Cladocera · Paleolimnology

1 Introduction

During the 20th century, many lakes surrounding Sudbury, Canada, became heavily acidified and metal-contaminated due to large-scale regional mining and smelting operations (Gunn 1996). This resulted in widespread damage to aquatic biota from multiple trophic levels and commonly included reduced species richness and, in some cases, extirpations (Keller et al. 2004). Beginning in the 1970s, there was a concerted effort to reduce the atmospheric output of SO_2 from local smelters by the installation of exhaust scrubbers and government-mandated emission reduction targets. These measures were largely successful at reducing acid deposition, and decades later, many Sudbury lakes experienced marked increases in pH (Keller et al. 1986, 1999a, 2007). This chemical recovery has stimulated interest in the evaluation of biological recovery of the aquatic ecosystems in the region (e.g. Yan et al. 1996; Keller and Yan 1998; Smol et al. 1998; Valois et al. 2010; Khan et al. 2012; Palmer et al. 2013).

Despite rising lakewater pH, aquatic communities (e.g. fish, invertebrates, and zooplankton) in many Sudbury lakes remain impoverished relative to non-acidified lakes throughout Ontario (Keller et al.

A. L. Labaj (✉) · A. Jeziorski · J. Kurek · J. P. Smol
Paleoecological Environmental Assessment and Research Laboratory (PEARL) Department of Biology, Queen's University,
Kingston, ON K7L 3N6, Canada
e-mail: a.labaj@queensu.ca

1999a). Within ~25 km of Sudbury, many of these recovering lakes continue to exceed provincial water-quality guidelines for copper and nickel concentrations, metals that can cause biological damage (Keller et al. 1999a; Valois et al. 2010). Even in lakes with water quality sufficient for zooplankton survival, species may be unable to recolonize due to lack of colonization vectors and factors reducing the viability of species resting eggs (Gray and Arnott 2009). Should recolonization be successful, altered food webs persist in previously acidified lakes long after they have recovered in pH, a phenomenon known as biotic resistance (Keller and Yan 1998; Gray and Arnott 2009; Valois et al. 2010; Jeziorski et al. 2013). For example, lake acidification may extirpate fish communities, transferring the role of top predators to invertebrates and, in turn, preventing the recovery (to a pre-impact state) of zooplankton communities (Yan et al. 1991; Gray and Arnott 2009). The biological recovery of acidified lakes may also be limited by regional changes in climate, such as increasing air temperatures and shifting ice phenology (Keller 2007; Futter 2003). Furthermore, due to the interactions of multiple environmental stressors, it may be unrealistic to expect a complete return of biological species to their pre-impact assemblage (Valois et al. 2011; Arseneau et al. 2011; Palmer et al. 2013). Although the trajectory of biological recovery can vary based on the additional stressors present, biota are generally expected to shift toward pre-impact states in time as acid stress is removed (Keller et al. 1999b).

Here, we use paleolimnology to examine the cladoceran assemblages of an extensively monitored lake that acidified severely and was subsequently limed and fertilized in the early 1970s. This indirect monitoring approach allows for the tracking of cladoceran assemblages before, during, and after acidification. The intermediate trophic position of cladocerans in lake food webs makes them sensitive to changes in both primary production (food supply) and predation by invertebrates and fish (Hessen et al. 1995). Cladoceran communities are also influenced by acidification to a large extent, and consequently some species are reliable indicators of acidity-related factors (Keller et al. 1990; Havens et al. 1993; Valois et al. 2010; Kurek et al. 2011). Specifically, we use the sedimentary cladoceran assemblages to (1) examine the cladoceran response to acidification and liming over the last ~250 years, (2) evaluate long-term trends in predation, and (3) assess the biological recovery of the Middle Lake cladoceran assemblage with respect to its pre-impact state (e.g. late 1800s).

2 Methods

2.1 Site Description

Middle Lake (N 46º 26.342′ W 81º 01.440′) is located within the City of Greater Sudbury, approximately 5 km SE of the Copper Cliff smelter (Fig. 1). The lake is underlain by quartzite (Scheider and Dillon 1976) and has a maximum depth and surface area of 15 m and 28.2 ha, respectively (Yan et al. 1996). Surrounding the lake are permanent homes, seasonal dwellings, and a road embankment at the southern end that was constructed in the 1970s. Middle Lake has been subject to prior paleolimnological investigation, using diatom fossils to reconstruct pH levels back to ~1900; however, this diatom-based reconstruction does not extend to pre-impact times (Dixit et al. 1990). From ~1900 to ~1940, diatom-inferred pH was stable at ~6; however, after this time, began to decline, and by the early 1970s, the lake had severely acidified to a pH of 4.3 (Dixit et al. 1990). In 1973, total Cu and Ni lakewater concentrations reached 496 $\mu g \cdot L^{-1}$ and 1,068 $\mu g \cdot L^{-1}$, respectively (Yan et al. 1996), well above the 5 $\mu g \cdot L^{-1}$ and 25 $\mu g \cdot L^{-1}$ provincial water-quality guidelines for these metals (Pearson et al. 2002). The history of planktivorous fish presence/absence in Middle Lake was also reconstructed using larval remains of the phantom midge *Chaoborus* spp. and likely consisted of one or two fish species, which were extirpated from the lake as pH dropped below ~6 in the 1940s (Uutala and Smol 1996). The lake was experimentally limed in 1973 (Scheider and Dillon 1976), and the pH of the lake has remained above 6 since this time (pH=7.7 in July 2012). In an attempt to remediate the surrounding land and promote plant re-growth, phosphorus treatments were applied to the watershed between 1975 and 1978 (Scheider and Dillon 1976; Yan and Lafrance 1984; Yan et al. 1996). Further liming and fertilization of the catchment was carried out in 1983–1984, maintaining the pH of the lake (Yan et al. 1996). Three species of fish (smallmouth bass, Iowa darters, and brook stickleback) were also stocked in the lake at this time; however, stocking was unsuccessful due to high metal levels in the water (specifically Cu, at 60 $\mu g \cdot L^{-1}$; Yan and Dillon 1984; Poulin et al. 1990; Girard et al. 2006). Metal levels have been in steady decline since

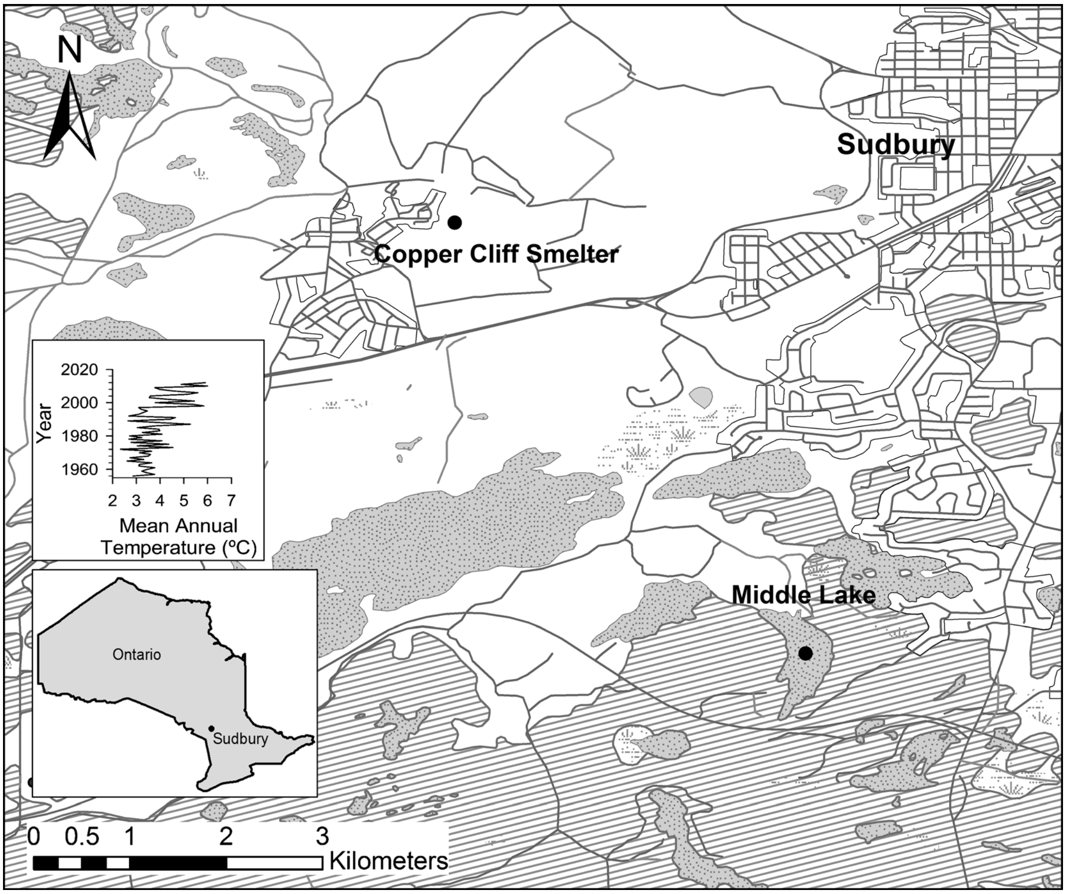

Fig. 1 Map of Sudbury showing the locations of Middle Lake and the Copper Cliff smelter. Hashmarks indicate vegetated areas. Inset: Mean annual air temperature (1956–2012) measured at Sudbury Airport (Climate station ID: 6068150). Data obtained from Environment Canada historical climate records (http://climate.weather.gc.ca/)

liming and emission reduction efforts were implemented, and Cu and Ni concentrations were 18 $\mu g \cdot L^{-1}$ and ~100 $\mu g \cdot L^{-1}$, respectively, as of 2007 (Palmer et al. 2013). As of 2008, the lake contains six fish species (yellow perch, smallmouth bass, brown bullhead, pumpkinseed, northern pike, and walleye; Leuk et al. 2010).

2.2 Field Methods

In July 2012, a sediment core was collected from the deepest basin of the lake at ~13 m water depth using a Glew (1989) gravity corer. The sediment core was sectioned in the field at 0.25-cm intervals from 0 to 15 cm, and 0.5-cm intervals beyond 15 cm using a core extruder (Glew 1988). Sediments were then stored in a cooler with icepacks until transportation to the laboratory.

2.3 Laboratory Methods

Gamma spectroscopy was used to determine sediment age, following the methods described by Schelske et al. (1994). ^{210}Pb and ^{137}Cs were measured from 18 sediment intervals. The Constant Rate of Supply (CRS) model described by Appleby (2001) was used to determine sediment ages based on unsupported ^{210}Pb concentrations. The ^{137}Cs peak, resulting from nuclear weapons testing at ~1963, was compared to the ^{210}Pb dates to verify their accuracy. Background ^{210}Pb levels were reached at 10.5–10.75 cm, corresponding to the ~1890s (Fig. 2). Sediment ages below 10.5–10.75 cm were determined by extrapolation using linear regression. The bottom of the core (25 cm) corresponds to ~1740 A.D. (Fig. 2). We note ^{137}Cs begins to decline at ~5.5 cm, indicating a date of ~1963 (Appleby 2001).

Past trends in sedimentary-inferred chlorophyll-*a* (Chl-*a*) were estimated using visible reflectance spectroscopy (VRS) techniques (Wolfe et al. 2006). Importantly, this method utilizes the spectral signatures of Chl-*a* and its degradation products between 650 and 700 nm wavelengths to infer Chl-*a* levels from sediments. The accuracy of the VRS method has been verified using paleolimnological studies of lakes with known primary production histories (Michelutti et al. 2010).

Cladoceran remains were prepared largely following the methods outlined by Korhola and Rautio (2001). Freeze–dried sediments were deflocculated by combining with a 10 % KOH solution and heating at ~80ºC for 20 min and then rinsed over a 38-μm mesh sieve and concentrated into a glass vial. Three drops of ethanol and safranin solution were added as a preservative and stain, respectively. Between three and eight 50-μL aliquots of the concentrated remains were applied to each slide, followed by glycerine jelly to mount a glass cover slip. Cladoceran remains (headshields, carapaces, etc.) were examined at 200× magnification under brightfield illumination on a Leica DMR microscope. The primary taxonomic resources used for the identification of cladoceran remains were Szeroczyńska and Sarmaja-Korjonen (2007) and Korosi and Smol (2012a, 2012b). Entire coverslips were scanned, and a minimum of 100 individuals were enumerated for each interval, as per counting effort guidelines of Kurek et al. (2010b).

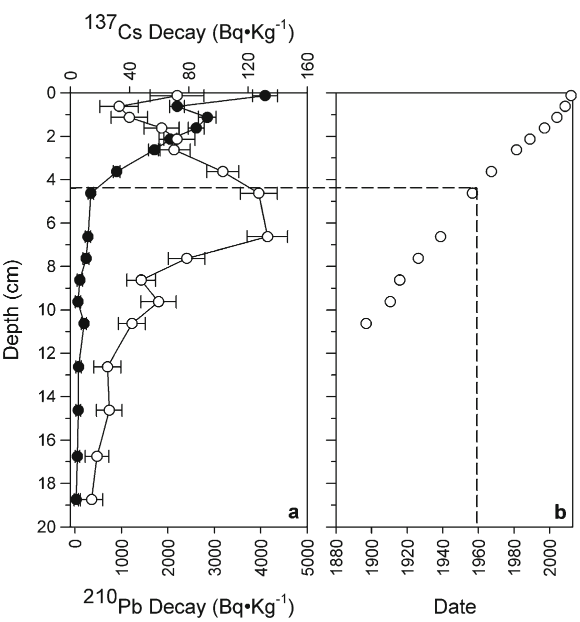

Fig. 2 **a** ^{210}Pb (*black circles*) and ^{137}Cs (*white circles*) decay versus interval depth. *Error bars* represent standard error. **b** Sediment date vs. depth based on ^{210}Pb spectroscopy. Beyond this depth, dates were calculated based on linear regression (age= −11.102 (interval depth) +2012.9)

Daphnia were grouped into two species complexes: the *Daphnia pulex* complex (possibly consisting of *D. pulex*, *Daphnia pulicaria*, *Daphnia catawba*, and *Daphnia minehaha*) and the *Daphnia longispina* complex (possibly consisting of *Daphnia ambigua*, *Daphnia mendotae*, *Daphnia dentifera*, *Daphnia dubia*, and *Daphnia longiremis*) (Hebert 1995), based on the occurrence of stout spines along the middle region of the postabdominal claw (Korosi and Smol 2012a). *Bosmina* spp. size attributes (i.e. carapace, mucro, and antennule lengths) were measured using Northern Eclipse image analysis software following the methods described in Korosi et al. (2010). To ensure meaningful size variations were detected, a minimum of 35 remains were measured for each *Bosmina* size attribute per sediment interval (Brahney et al. 2010).

3 Results

3.1 Cladoceran Assemblages

A total of 29 taxa were recovered from Middle Lake (Fig. 3). Sediment intervals in the pre-impact period (~1740 to ~1890; 25 cm–10.5 cm) had an average species richness (rarefied) of ~11. This was considerably higher than the average species richness (rarefied) of ~4 during the acidification (1890 to early 1970s; 9.75 to 4 cm) and post-liming (early 1970s to 2012; 3.75 cm to sediment surface) periods. Four taxa comprised >87 % of the assemblage throughout all sediment intervals: *Chydorus brevilabris*, *Bosmina* spp. (primarily *Bosmina longirostris*), *D. pulex* complex, and *D. longispina* complex. All other taxa did not exceed 2 % in 2 or more sediment intervals and were therefore considered rare. Several species, including *Alona intermedia* and *Holopedium glacialis*, were present in the lake during the pre-impact period but disappeared by the late 1800s. *Acroperus harpae*, *Alonella excisa*, *Alonella nana*, and *Leptodora kindtii* disappeared from the lake during the acidification period.

During the pre-impact period, the common cladoceran taxa experienced only minor changes, with *Bosmina* spp. and *C. brevilabris* averaging 77 % and 3 % relative abundance, respectively (Fig. 3). The *D. pulex* complex, initially present at 17 %, declined to 9 % relative

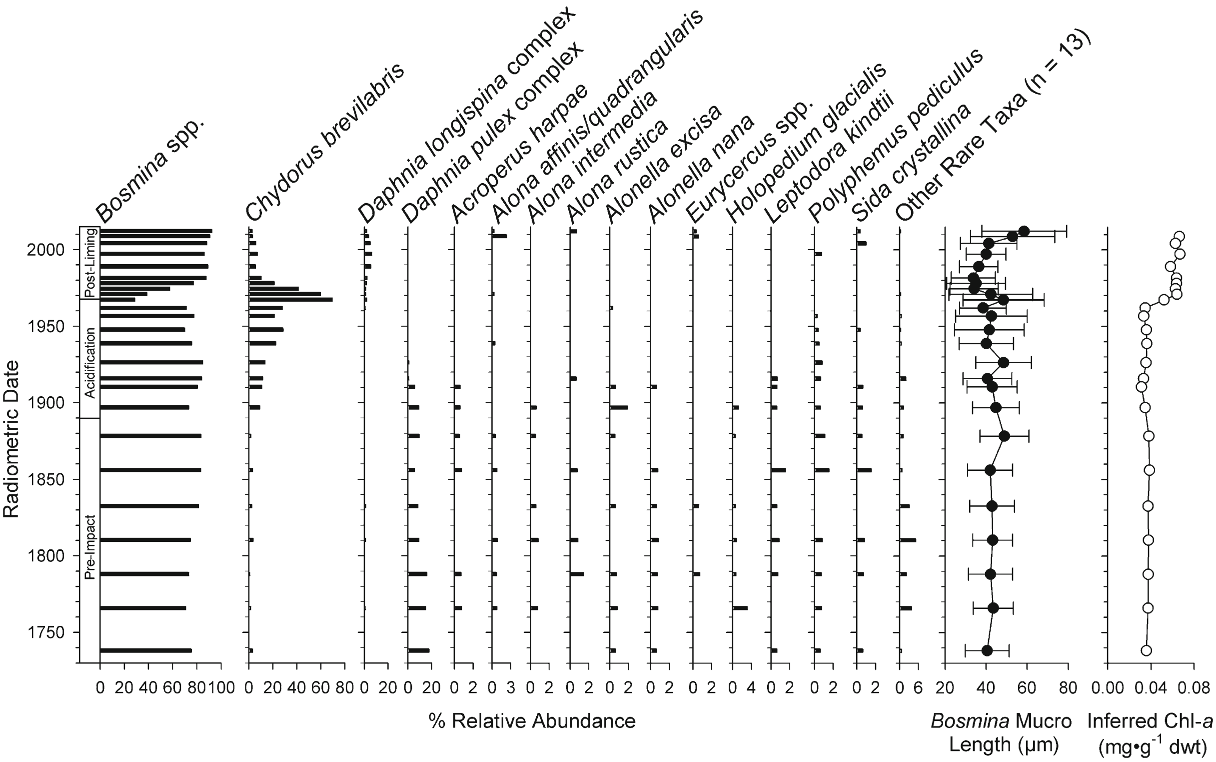

Fig. 3 Percent relative abundance of select taxa present in Middle Lake. Average *Bosmina* mucro length (error bars represent standard error) and VRS chlorophyll-*a* are indicated for each interval. ^{210}Pb dates are shown to the left

abundance by the 1890s. During the acidification period, *C. brevilabris* increased to approximately 30 % relative abundance by the late 1950s, while the *D. pulex* complex disappeared from the record in ~1940, replaced by the *D. longispina* complex. Approximately consistent with liming, *C. brevilabris* relative abundance rapidly increased to ~70 %, followed by a rapid decline to ~5 % relative abundance by 1990, where it persisted for the rest of the core, along with *Bosmina* at ~90 % relative abundance, and the *D. longispina* complex at 2–6 % relative abundance.

Bosmina antennule and carapace lengths showed little change throughout the record, remaining at an average of ~93 μm and ~215 μm, respectively. Mucro length was stable at an average of ~43 μm throughout the pre-impact and acidification periods, but began to decline around the time of liming, reaching a minimum of ~33 μm by ~1980 (Fig. 3). Mean mucro length then increased throughout the remainder of the post-liming period, reaching ~60 μm by 2012 (Fig. 3). The mean mucro length time series was not significant across the record (Mann–Kendall trend test $\tau=0.15$, $p=0.29$) nor during the post-liming period ($\tau=-0.33$, $p=0.21$).

3.2 VRS Chlorophyll-*a*

VRS Chl-*a* levels remained stable during the pre-impact period [~0.035 mg·g^{-1} dry weight (dwt), Fig. 3]. Levels remained consistently low and stable throughout the acidification period (Fig. 3). Coincident with the onset of lake fertilization, Chl-*a* increased sharply to a level of ~0.064 mg·g^{-1} dwt and has since remained at this level (Fig. 3).

4 Discussion

4.1 Cladoceran Assemblage Response to Acidification and Liming

The cladoceran assemblage of Middle Lake has undergone marked changes during the past ~130 years, largely tracking the impact of anthropogenic acidification.

C. brevilabris was present throughout the sediment record; however, beginning in the late 1800s, there was an increase in its relative abundance, coincident with the rise of development in the Sudbury region and the use of open-pit ore roasting beds (Gunn et al. 1995; Pearson et al. 2002). The monitoring record of Yan et al. (1996) extends only a few months prior to liming and thus did not detect the long-term rise in *C. brevilabris* observed in the paleolimnological record. Similar long-term increases in *C. brevilabris* have been observed in other acid-stressed systems, such as several heavily acidified lakes downwind of an iron sintering plant near Wawa, Ontario (Jeziorski et al. 2013). Due to its high survival under low-pH conditions, *C. brevilabris* is able to take advantage of the reduced competition in acidified lakes, thus often reaching relatively high abundances in these systems (Kurek et al. 2011; Jeziorski et al. 2013). *C. brevilabris* reached its maximum relative abundance during peak acidification of the lake in the early 1970s, subsequently experiencing a rapid decline around the time of liming.

Throughout the pre-impact and acidification periods, VRS Chl-*a* remained stable, an indication that acidification had little effect on the primary production of the lake. The VRS Chl-a measurements (Fig. 3) appear to track the fertilization of Middle Lake well. We interpret the step-like increase in VRS Chl-*a* to be a result of the lake fertilization that began in 1975. Following fertilization, phytoplankton standing stocks increased by ~400 %, in addition to a shift from chrysophytes to blue–green algae as the dominant summer phytoplankton taxa (Scheider and Dillon 1976; Yan and Lafrance 1984). Because our methodology also measures chlorophyll-a degradation products, this increase cannot be explained by diagenetic processes (Michelutti et al. 2010). According to our age-depth model, the rapid increase in VRS Chl-a appears a few years before fertilization occurred in Middle Lake, indicating that our linear-interpolated model is overestimating sediment age at this point in our record (Fig. 3). Nonetheless, the most likely explanation of the increase in primary production is related to fertilization treatments to the catchment of Middle Lake between 1975 and 1978. The final fertilization of the lake took place in 1984; however, VRS Chl-*a* levels have remained elevated from pre-impact levels, possibly reflecting longer ice-free seasons (Futter 2003) and significant increased average air temperatures in this region in recent decades (Mann–Kendall trend test τ=0.34, p=0.0001; Fig. 1).

4.2 Biological Recovery of Cladocerans

For the past ~250 years, the cladoceran record from Middle Lake has been dominated by *Bosmina* spp. and, to some extent, *C. brevilabri*s. Daphniids from the *D. pulex* complex comprised up to 15 % of the assemblage during the pre-impact period, however declined with acidification and were completely replaced by the *D. longispina* complex (likely *D. mendotae*; Yan et al. 2004), which now inhabits the lake. Furthermore, many rare taxa that disappeared during the acidification period have not yet re-colonized the lake, even though Middle Lake has had a pH>6.5 since 1974. Moreover, the cladoceran community (sampled by net hauls through the water column) remains different from reference lakes in the region (Yan et al. 2004; Palmer et al. 2013). In marked contrast, the copepod community of Middle Lake is considered fully recovered (Palmer et al. 2013). Such incomplete recovery of cladocerans relative to other biological groups is a trend that has been observed in other acid-impacted regions of eastern North America. For example, Arseneau et al. (2011) observed shifts in diatom and chrysophyte species indicative of biological recovery in Adirondack (USA) lakes; however, cladoceran species failed to show a response to rising lakewater pH. It is apparent that additional factors continue to prevent the complete recovery of the cladoceran community in Middle Lake. Although pH has recovered, the influence of multiple stressors has likely changed Middle Lake from its pre-impact conditions in ways beyond those related to industrial emissions. Construction of the highway embankment at the southern end of the lake has resulted in steady increases in salinity (Girard et al. 2006). Furthermore, shoreline development and invasion of Eurasian water milfoil in many lakes of the region have the potential to alter littoral habitat. Despite these possible factors, the cladoceran assemblage has changed little in the past 30 years, suggesting that the above stressors have had little impact. The most obvious factors limiting biological recovery include continued metal contamination (Valois et al. 2010; Webster et al. 2013), dispersal/recolonization limitations (Gray and Arnott 2009), altered predation dynamics (Keller and Yan 1998; Webster et al. 2013), and a changing climate (Futter 2003; Keller 2007).

Biotic recovery may be hindered by the inability of organisms to disperse and successfully recolonize lakes that have recovered from ecological degradation. Cladocera typically recolonize a lake from two sources: the resting egg bank contained within the lake sediments and via transportation vectors (e.g. streams, animal movements, and wind) from nearby lakes (Keller and Yan 1998; Gray and Arnott 2009). We note several rare taxa, including *Alona affinis/quadrangularis*, *Alona rustica*, *Eurycercus* spp., *Sida crystallina*, and *Polyphemus pediculus*, appeared during the post-liming phase, after being absent from the fossil record for at least several years prior to liming, suggesting that they disappeared from the lake and returned. Similarly, in the net-haul record, Yan et al. (2004) observed that five cladoceran species arrived in Middle Lake within a 1-year period yet none persisted. Collectively, direct monitoring and the sediment record indicate that cladoceran recovery in Middle Lake is not being limited by dispersal. Likely, environmental conditions and biotic processes have greater relative importance compared to dispersal-related factors in structuring fossil cladoceran communities (Kurek et al. 2011).

An additional potential barrier to recovery is biotic resistance. In acidifying lakes, fish are often eliminated after pH drops below 6 (Beamish and Harvey 1972), while invertebrates such as *Chaoborus* can survive pH as low as 3.5 (Havas and Likens 1985). In the absence of fish, *Chaoborus* may fill the role as dominant predators (Appelberg et al. 1993; Labaj et al. 2013a), reducing the density of small-bodied zooplankton species (Vanni 1988; Wellborn et al. 1996). Within Middle Lake, Uutala and Smol (1996) noted that *Chaoborus americanus*, an indicator of fishless conditions or low fish planktivory (Sweetman and Smol 2006; Kurek et al. 2010a), began increasing in the 1950s during acidification. *Bosmina* spp. can modify body size attributes in response to differing predation types, with longer morphs typically reflecting greater invertebrate predation and shorter morphs reflecting reduced invertebrate predation (possibly due to greater planktivourus fish predation) (Korosi et al. 2013; Labaj et al. 2013b). We noted that although *Bosmina* mucro length experienced a decline and subsequent increase in recent decades, no significant trends were observed over the entire record or post-liming, suggesting that invertebrate predation has remained stable. Since ~1990, Middle Lake has contained a large population of yellow perch (*Perca flavescens*), a species that consumes zooplankton in its juvenile stage and later life stages if sufficient benthic invertebrates are unavailable (Poulin et al. 1990; Leuk et al. 2010; Valois et al. 2010). Predation by yellow perch may continue to present a barrier to cladoceran recovery in Middle Lake (Leuk et al. 2010), especially as metal levels continue to decline (Webster et al. 2013).

Although the effects of acidification from a lakewater pH perspective have been resolved in Middle Lake, it is not surprising that the cladoceran assemblage has failed to return to its pre-impact state (species richness and composition). The most likely impediment to full biological recovery is that multiple stressors continue to affect the lake's biota, in addition to physical/chemical properties. Cu and Ni contamination, though decreasing since emission reductions were implemented, may continue to limit the survival of sensitive taxa, including daphniids (Yan et al. 2004; Khan et al. 2012; Webster et al. 2013). Additionally, Middle Lake has experienced a shift in stratification patterns, switching from weak to strong stratification in 1974 (following liming and fertilization; Girard et al. 2006). Such shifts in thermal regime have been observed in lakes throughout the Sudbury region, due to increasing DOC (Girard et al. 2006), forest re-growth (Tanentzap et al. 2008), and significant warming regional temperatures (Fig. 1). Shifts in lake stratification may have possible implications to cladoceran communities, especially small-bodied zooplankton such as *Bosmina* (Moore et al. 1995). Changing lake stratification patterns may also affect invertebrate predation upon pelagic cladocerans such as the bosminids (Labaj et al. 2013b). As these stressors continue to impact aquatic ecosystems throughout the Boreal Shield ecozone, it is increasingly unlikely that complete biotic recovery will occur in previously acidified systems, and our ability to predict and manage ecosystem response may be challenged.

5 Conclusions

Although the restoration efforts in Middle Lake have raised and maintained the pH at circumneutral levels since 1973, a legacy of acid and metal damage is still evident, and the modern cladoceran assemblage is impoverished relative to its pre-impact state. Our long-term record indicates that biological impacts associated with industrial development in Sudbury began as early as the ~1890s and increased in severity until the lake was limed in 1973. *C. brevilabris* tracked the acidification history of Middle Lake, reaching maximum relative abundance at peak acidification and rapidly declining with approximately similar timing to liming. Many species present prior to the onset of acidification are still missing from the lake, and the dominant *Daphnia* species differ between contemporary and pre-impact periods. Recovery of the cladoceran assemblage is likely still limited by high levels of Cu and Ni in the water and sediments, increased predation by juvenile yellow perch in the littoral zone, enhanced predation by invertebrates in the pelagic zone, and regional climate change. Although great strides have been made toward pH recovery in the Sudbury region, further research into the effects of multiple stressors on biological recovery is warranted if we are to fully appreciate ecosystem trajectory following the widespread and often severe impacts of human activities on aquatic ecosystems during the 20th century.

Acknowledgments This study was funded by a Natural Sciences and Engineering Research Council of Canada (NSERC) grant to John Smol and NSERC graduate scholarship to Andrew Labaj. We thank Joshua Thienpont for his assistance in the field, and the Cooperative Freshwater Ecology Unit, Laurentian University, Sudbury, for providing chemical monitoring data of Middle Lake. Additionally, we wish to thank Edwin Morelli for assistance with chlorophyll-*a* analysis, Clare Nelligan for assistance with creating a map, and an anonymous reviewer for comments that greatly strengthened our manuscript.

References

Appelberg, M., Henrikson, B.-I., Henrikson, L., & Svedäng, M. (1993). Biotic interactions within the littoral community of Swedish forest lakes during acidification. *Ambio, 22*, 290–297.

Appleby, P. G. (2001). Chronostratigraphic techniques in recent sediments. In W. M. Last & J. P. Smol (Eds.), *Tracking environmental change using lake sediments. 1. Basin analysis, coring, and chronological techniques* (pp. 171–203). Dordrecht: Springer.

Arseneau, K. M. A., Driscoll, C. T., Brager, L. M., Ross, K. A., & Cumming, B. F. (2011). Recent evidence of biological recovery from acidification in the Adirondacks (New York, USA): a multiproxy paleolimnological investigation of Big Moose Lake. *Canadian Journal of Fisheries and Aquatic Sciences, 68*, 575–592.

Beamish, R. J., & Harvey, H. H. (1972). Acidification of the La Cloche Mountain lakes, Ontario, and resulting fish mortalities. *Journal of the Fisheries Research Board of Canada, 29*, 1131–1143.

Brahney, J., Routledge, R., Bos, D. G., & Pellatt, M. G. (2010). Changes to the productivity and trophic structure of a sockeye salmon rearing lake in British Columbia. *North American Journal of Fisheries Management, 30*, 433–444.

Dixit, S. S., Dixit, A. S., & Smol, J. P. (1990). Paleolimnological investigation of three manipulated lakes from Sudbury, Canada. *Hydrobiologia, 214*, 245–252.

Futter, M. N. (2003). Patterns and trends in Southern Ontario lake ice phenology. *Environmental Monitoring and Assessment, 88*, 431–444.

Girard, R., Yan, N. D., Heneberry, J., & Keller, W. (2006). Physical and chemical data series from Clearwater, Lohi, Middle and Hannah lakes near Sudbury, Ontario: long-term responses to liming, and natural recovery from historical acidification and metal contamination. Ontario Ministry of Environment Report #5429e.

Glew, J. R. (1988). A portable extruding device for close interval sectioning of unconsolidated core samples. *Journal of Paleolimnology, 1*, 235–239.

Glew, J. R. (1989). A new trigger mechanism for sediment samplers. *Journal of Paleolimnology, 2*, 241–243.

Gray, D. K., & Arnott, S. E. (2009). Recovery of acid damaged zooplankton communities: measurement, extent, and limiting factors. *Environmental Reviews, 17*, 81–99.

Gunn, J. M. (1996). Restoring the smelter-damaged landscape near Sudbury, Canada. *Restoration and Management Notes, 14*, 129–136.

Gunn, J. M., Keller, W., Negusanti, J., Potvin, R., Beckett, P., & Winterhalder, K. (1995). Ecosystem recovery after emission reductions: Sudbury, Canada. *Water, Air, and Soil Pollution, 85*, 1783–1788.

Havas, M., & Likens, G. E. (1985). Toxicity of aluminum and hydrogen ions to *Daphnia catawba*, *Holopedium gibberum*, *Chaoborus puntcipennis*, and *Chironomus anthrocinus* from Mirror Lake, New Hampshire. *Canadian Journal of Zoology, 63*, 1114–1119.

Havens, K. E., Yan, N. D., & Keller, W. (1993). Lake acidification: effects on crustacean zooplankton populations. *Environmental Science & Technology, 27*, 1621–1624.

Hebert, P. D. N. (1995). The Daphnia of North America: an illustrated fauna. CD-ROM, Guelph: University of Guelph.

Hessen, D. O., Faafeng, B. A., & Anderson, T. (1995). Replacement of herbivore zooplankton species along gradients of ecosystem productivity and fish predation pressure. *Canadian Journal of Fisheries and Aquatic Sciences, 52*, 733–742.

Jeziorski, A., Keller, W., Paterson, A. M., Greenaway, C. M., & Smol, J. P. (2013). Aquatic ecosystem responses to rapid recovery from extreme acidification and metal contamination in lakes near Wawa, Ontario. *Ecosystems, 16,* 209–223.

Keller, W. (2007). Implications of climate warming for Boreal Shield lakes: a review and synthesis. *Environmental Reviews, 15*, 99–112.

Keller, W., & Yan, N. D. (1998). Biological recovery from acidification: zooplankton communities as a model of patterns and processes. *Restoration Ecology, 6*, 364–375.

Keller, W., Pitblado, J. R., & Conroy, N. I. (1986). Water quality improvements in the Sudbury, Ontario, Canada area related to reduced smelter emissions. *Water, Air, and Soil Pollution, 31*, 765–774.

Keller, W., Yan, N. D., Holtze, K. E., & Pitblado, J. R. (1990). Inferred effects of lake acidification on *Daphnia galeata mendotae. Environmental Science & Technology, 24*, 1259–1261.

Keller, W., Gunn, J. M., & Yan, N. D. (1999a). Acid rain-perspectives on lake recovery. *Journal of Aquatic Ecosystem Stress and Recovery, 6*, 207–216.

Keller, W., Heneberry, J. H., & Gunn, J. M. (1999b). Effects of emission reductions from the Sudbury smelters on the recovery of acid- and metal-damaged lakes. *Journal of Aquatic Ecosystem Stress and Recovery, 6*, 198–198.

Keller, W., Heneberry, J., Gunn, J. M., Snucins, E., Morgan, G., & Leduc, J. (2004). Recovery of acid and metal-damaged lakes near Sudbury Ontario: trends and status. Ecological Risk Assessment, Sudbury Soils Study 53.

Keller, W., Yan, N. D., Gunn, J. M., & Heneberry, J. (2007). Recovery of acidified lakes: lessons from Sudbury, Ontario, Canada. *Water, Air, and Soil Pollution: Focus, 7*, 317–322.

Khan, F. R., Keller, W., Yan, N. D., Welsh, P. G., Wood, C. M., & McGreer, J. C. (2012). Application of biotic ligand and toxic unit modeling approaches to predict improvements in zooplankton species richness in smelter-damaged lakes near Sudbury, Ontario. *Environmental Science & Technology, 46*, 1641–1649.

Korhola, A., & Rautio, M. (2001). Cladocera and other brachiopod crustaceans. In J. P. Smol, H. J. B. Birks, & W. M. Last (Eds.), *Tracking environmental change using lake sediments. 4. Zoological indicators* (pp. 5–41). Dordrecht: Springer.

Korosi, J. B., & Smol, J. P. (2012a). An illustrated guide to the identification of cladoceran subfossils from lake sediments in northeastern North America: part 1–the Daphniidae, Leptodoridae, Bosminidae, Polyphemidae, Holopedidae, Sididae, and Macrothricidae. *Journal of Paleolimnology, 48*, 571–586.

Korosi, J. B., & Smol, J. P. (2012b). An illustrated guide to the identification of cladoceran subfossils from lake sediments in northeastern North America: part 2–the Chydoridae. *Journal of Paleolimnology, 48*, 587–622.

Korosi, J. B., Paterson, A. M., DeSellas, A. M., & Smol, J. P. (2010). A comparison of pre-industrial and present-day changes in *Bosmina* and *Daphnia* size structure from soft-water Ontario lakes. *Canadian Journal of Fisheries and Aquatic Sciences, 67*, 754–762.

Korosi, J. B., Kurek, J., & Smol, J. P. (2013). A review on utilizing *Bosmina* size structure archived in lake sediments to infer historic shifts in predation regimes. *Journal of Plankton Research, 35*, 444–460.

Kurek, J., Cwynar, L. C., Weeber, R. C., Jefferies, D. S., & Smol, J. P. (2010a). Ecological distributions of Chaoborus species in small, shallow lakes from the Canadian Boreal Shield ecozone. *Hydrobiologia, 652*, 207–221.

Kurek, J., Korosi, J. B., Jeziorski, A., & Smol, J. P. (2010b). Establishing reliable minimum count sizes for cladoceran subfossils sampled from lake sediments. *Journal of Paleolimnology, 44*, 603–612.

Kurek, J., Weeber, R. C., & Smol, J. P. (2011). Environment trumps predation and spatial factors in structuring cladoceran communities from Boreal Shield lakes. *Canadian Journal of Fisheries and Aquatic Sciences, 68*, 1408–1419.

Labaj, A. L., Kurek, J., & Smol, J. P. (2013a). *Chaoborus americanus* predation influences *Bosmina* mucro lengths in fishless lakes. *Journal of Paleolimnology*. doi:10.1007/s10933-013-9751-5.

Labaj, A. L., Kurek, J., Weeber, R. C., & Smol, J. P. (2013b). Long-term changes in invertebrate size structure and composition in a boreal headwater lake with a known minnow introduction. *Journal of Limnology, 72*, 215–226.

Leuk, A., Morgan, G. E., Wissel, B., Gunn, J. M., & Ramcharan, C. W. (2010). Rapid and unexpected effects of piscivore introduction on trophic position and diet of perch (*Perca flavescens*) in lakes recovering from acidification and metal contamination. *Freshwater Biology, 55*, 1616–1627.

Michelutti, N., Blais, J. M., Cumming, B. F., Paterson, A. M., Rühland, K. M., Wolfe, A. P., et al. (2010). Do spectrally inferred determinations of chlorophyll *a* reflect trends in lake trophic status? *Journal of Paleolimnology, 43*, 205–217.

Moore, M. V., Folt, C. L., & Stemberger, R. S. (1995). Consequences of elevated temperatures for zooplankton assemblages in temperate lakes. *Archiv für Hydrobiologie, 135*, 289–319.

Palmer, M. E., Keller, W., & Yan, N. D. (2013). Gauging recovery of zooplankton from historical acid and metal contamination: the influence of temporal changes in restoration targets. *Journal of Applied Ecology, 50*, 107–118.

Pearson, D. A. B., Gunn, J. M., & Keller, W. (2002). The past, present, and future of Sudbury's lakes. In D. H. Rousell and K. Jansons (Eds.). Physical Environment of the Sudbury Region, Ontario Geographical Survey Special Volume 6 (pp. 195–215).

Poulin, D. J., Gunn, J. M., & Laws, K. M. (1990). Fish species present in Sudbury lakes–results of the 1989 and 1990 urban lakes survey. Ontario Ministry of Natural Resources, Sudbury District.

Scheider, W., & Dillon, P. J. (1976). Neutralization and fertilization of acidified lakes near Sudbury, Ontario. Proceedings of the 11th Canadian Symposium, Water Pollution Research Canada (pp. 93–100).

Schelske, C. L., Peplow, A., Brenner, M., & Spence, C. N. (1994). Low-background gamma counting applications for ^{210}Pb dating of sediments. *Journal of Paleolimnology, 10*, 115–128.

Smol, J. P., Cumming, B. F., Dixit, A. S., & Dixit, S. S. (1998). Tracking recovery patterns in acidified lakes: a paleolimnological perspective. *Restoration Ecology, 6*, 318–326.

Sweetman, J. N., & Smol, J. P. (2006). Reconstructing fish populations using *Chaoborus* (Diptera: Chaoboridae) remains—a review. *Quaternary Science Reviews, 25*, 2013–2023.

Szeroczyńska, K., & Sarmaja-Korjonen, K. (2007). *Atlas of subfossil Cladocera from central and northern Europe*. Świecie: Friends of the Lower Vistula Society.

Tanentzap, A. J., Yan, N. D., Keller, B., Girard, R., Haneberry, J., Gunn, J. M., et al. (2008). Cooling lakes while the world warms: effects of forest regrowth and increased dissolved organic matter on the thermal regime of a temperate, urban lake. *Limnology and Oceanography, 53*, 404–410.

Uutala, A. J., & Smol, J. P. (1996). Paleolimnological reconstructions of long-term changes in fisheries status in Sudbury area lakes. *Canadian Journal of Fisheries and Aquatic Sciences, 53*, 174–180.

Valois, A., Keller, W., & Ramcharan, C. W. (2010). Abiotic and biotic processes in lakes recovering from acidification: the relative roles of metal toxicity and fish predation as barriers to zooplankton reestablishment. *Freshwater Biology, 55*, 2585–2597.

Valois, A., Keller, W., & Ramcharan, C. W. (2011). Recovery in a multiple stressor environment: using the reference condition approach to examine zooplankton community change along opposing gradients. *Journal of Plankton Research, 33*, 1417–1429.

Vanni, M. J. (1988). Freshwater zooplankton community structure: introduction of large invertebrate predators and large herbivores to a small-species community. *Canadian Journal of Fisheries and Aquatic Sciences, 45*, 1758–1770.

Webster, N. I., Keller, W., & Ramcharan, C. W. (2013). Restoration of zooplankton communities in industrially damaged lakes: influences of residual metal contamination and the recovery of fish communities. *Restoration Ecology, 21*, 785–792.

Wellborn, G. A., Skelly, D. K., & Werner, E. E. (1996). Mechanisms creating community structure across a freshwater habitat gradient. *Annual Reviews of Ecology, Evolution, and Systematics, 27*, 337–363.

Wolfe, A. P., Vinebrooke, R. D., Michelutti, N., Rivard, B., & Das, B. (2006). Experimental calibration of lake-sediment spectral reflectance to chlorophyll *a* concentrations: methodology and paleolimnological validation. *Journal of Paleolimnology, 36*, 91–100.

Yan, N. D., & Dillon, P. J. (1984). Experimental neutralization of lakes near Sudbury, Ontario. In J. Nriagu (Ed.), *Advances in environmental sciences series* (pp. 417–456). New Jersey: John Wiley and Sons.

Yan, N. D., & Lafrance, C. (1984). Response of acidic and neutralized lakes near Sudbury, Ontario, to nutrient enrichment. In J. Nriagu (Ed.), *Advances in environmental sciences series* (pp. 457–521). New Jersey: John Wiley and Sons.

Yan, N. D., Keller, W., MacIsaac, H. J., & McEachern, L. J. (1991). Regulation of zooplankton community structure of an acidified lake by *Chaoborus*. *Ecological Applications, 1*, 52–65.

Yan, N. D., Keller, W., Somers, K. M., Pawson, T. W., & Girard, R. E. (1996). Recovery of crustacean zooplankton communities from acid and metal contamination: comparing manipulated and reference lakes. *Canadian Journal of Fisheries and Aquatic Sciences, 53*, 1301–1327.

Yan, N. D., Girard, R., Heneberry, J. H., Keller, W., Gunn, J. M., & Dillon, P. J. (2004). Recovery of copepod, but not cladoceran, zooplankton from severe and chronic effects of multiple stressors. *Ecology Letters, 7*, 452–460.